

Reducing Emergency Medical Service response time via the reallocation of ambulance bases

L. C. Nogueira Jr · L. R. Pinto · P. M. S. Silva

Received: 22 November 2013 / Accepted: 25 March 2014 / Published online: 18 April 2014
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Abstract The demand for highly efficient and effective services and consumer goods is an essential prerequisite for modern organizations. In healthcare, efficiency and effectiveness mean reducing disabilities and maintaining human life. One challenge is guaranteeing rapid Emergency Medical Service (EMS) response. This study analyzes the EMS of Belo Horizonte, Brazil, using two modeling techniques: optimization and simulation. The optimization model locates ambulance bases and allocates ambulances to those bases. A simulation of this proposed configuration is run to analyze the dynamic behavior of the system. The main assumption is that optimizing the ambulance base locations can improve the system response time. Feasible solutions were found and the current system may be improved while considering economic and operational changes.

Keywords Facility location · Discrete event simulation · Optimization · Mathematical modeling

MSC codes 90B90

1 Introduction

The healthcare industry has significantly developed both technologically and scientifically over the last century

[1]. Diseases that killed millions of people in the XIX century may now be gone. However, population longevity, urban violence, the number of sedentary people, and the occurrence of aging diseases have all increased [2, 3]. Therefore, the demand for EMS in large cities is increasing at the same rate [3]. Thus, healthcare policies should be updated to keep up with these changes. In developing countries, the impact of such changes seems stronger, and new policies should be implemented as soon as possible.

PAHO (Pan American Health Organization [4]) noted that contagious infectious diseases represent 10 % of deaths in Brazil, while “modern diseases” such as heart diseases continually increase. Over the last 50 years, Brazil has changed mortality patterns from young people to the more complex and expensive diseases of old age [3].

The Brazil Health Ministry [5] created the SAMU, a Portuguese Acronym for Emergency Medical Service, in 2003. Physicians, nurses and assistants compose service teams that rescue victims.

The main function of this system is to treat patients from the moment of first call until their delivery to a hospital using a minimum of resources [6–8].

This study covers the EMS of Belo Horizonte. It suggests the average response time (21 min in 2010 [6]) can be partially explained by inefficient base locations. Currently, the system has 22 bases and 27 ambulances.

An optimization model for minimizing the response time (RT) was proposed and some experiments were performed to determine better ambulance base locations. The new base locations were input in to a discrete event simulation (DES) model to observe the dynamic behavior of the system and more accurately estimate the response time.

L. C. Nogueira Jr (✉) · L. R. Pinto · P. M. S. Silva
Federal University of Minas Gerais, Antônio Carlos, 6627,
Pampulha, Belo Horizonte, MG, Brazil
e-mail: luiznogueira@ufmg.br

L. R. Pinto
e-mail: luiz@dep.ufmg.br

P. M. S. Silva
e-mail: pedromarinho84@ibest.com.br

2 The operational research applied to ambulance systems

2.1 Location problems

Generally, facility location problems propose the best facility locations that match a geographically distributed demand (customers) subject to some constraints. Owen and Daskin [9] divided the location models into the three categories shown in Table 1.

Sahin and Sural [10] hierarchically reviewed the most common location systems, such as healthcare systems, solid waste management systems, production-distribution systems, education systems, telecommunications networks and EMS systems. Many others studies have focused on facility location problems for healthcare (Cooper [11]; Drezner [12]; Borrás and Pastor [13]; Snyder [14]).

2.2 Location problems applied to healthcare systems

Drezner [12] presented two strategic questions for healthcare systems in large cities concerning using minimal resources to attend to people's needs and the EMS size required for a suitable response time.

The Goldberg [15] state healthcare has been an important subject for operational research (OR) studies since 1960. This subject involves relevant decisions regarding, for example, base locations, dynamic ambulance positioning, vehicle dispatching, fleet sizing, and resource management.

Brotcome et al. [16] performed a detailed three-decade review on ambulance location models. This study summarizes probabilistic and deterministic models and their primary applications.

Many other studies relate to ambulance models. Berlin and Liebman [17] modeled EMS using two models. They used an optimization model to solve the facility location problem and a simulation model to solve the vehicle allocation problem. Mandell [18] used a queue model to study the covering problem. Rajagopalan et al. [19] developed a model to optimize the number of ambulances for a seasonal demand. This model determined the optimum number and location for ambulances during each time range. Tavakoli and Lightner [20] proposed a model to locate facilities and allocate ambulances in Fayetteville, NC in the USA. The objective was to increase the number of calls answered within 8 min.

Jardim et al. [21] developed a simplified optimization model to propose new ambulance base locations in Belo Horizonte, Brazil. This model aimed to reduce the system response time by proposing new ambulance base locations.

Iannoni et al. [22] and Iannoni and Morabito [23] proposed hypercube queueing models to locate ambulance bases on Brazilian highways. Morabito et al. [24] also proposed queueing models to analyze non-homogeneous servers in the EMS of Campinas, Brazil and along a Brazilian highway. Takeda et al. [7] used a similar methodology to analyze the centralization of EMS in the same city.

2.3 Discrete event simulations applied to healthcare systems

Discrete event simulations (DES) are a powerful tool to evaluate and analyze both new projects and existing systems that became more popular after the computer revolution [25, 26].

The sequence of events comprising an ambulance system begins after an accident (traffic accidents, natural hazards,) and finishes once the patient is delivered to a hospital [27]. Therefore, these systems are complex and dynamic. EMSs involve many processes and random events that make solving problems using analytical models difficult. These features make DES an attractive tool for analyzing these systems [28].

The response time, i.e., the elapsed time between notification of an occurrence and the ambulance arrival at the scene, is the main EMS performance measurement [29]. The World Health Organization (WHO) recommends an ideal response time of less than 8 min [30].

The average response time varies from one city to another. Al-Ghamdi [31] determined 10.23 min for Riyadh (Saudi Arabia); Wu and Hwang [32] determined 9 min for Tainan (Taiwan); Takeda [29] determined 10.94 min for Campinas (Brazil) and Silva and Pinto [6] determined 21.21 min for Belo Horizonte (Brazil).

3 Methodology

3.1 Database analysis

Belo Horizonte has approximately 2.5 million inhabitants in an area of approximately 334 km² [3]. The

Table 1 Main facility location problems

Deterministic location problems	Dynamic location problems	Stochastic location problems
p-median problems	Dynamic single facility location models	Probabilistic models
Covering problems	Dynamic multiple facility location models	Scenario planning models
Center problems	Alternative dynamic approaches	

ambulance system currently has one call center for the entire city, and this study analyzed 70,706 ambulance dispatches from 01/01/2008 to 09/13/2009. Figure 1 shows the distribution of calls in the city. For practical purposes, the city was divided into 71 regions and the calls were spread among these regions.

The call times were arranged into four periods of the day: i) dawn (12:00 AM–06:00 AM), ii) morning (06:00 AM–12:00 PM), iii) afternoon (12:00 PM–06:00 PM), and iv) evening (06:00 PM–12:00 AM). Table 2 shows the arrival rates.

This division is important because of daily seasonality. Another important factor is the average ambulance speed. The model supposes 55 km/h from midnight to 6:00 AM, 32 km/h from 6:00 AM to 6:00 PM and 42 km/h from 6:00 PM to midnight (based on EMS practices in Belo Horizonte).

3.2 Ambulance base locations

There are 22 ambulance bases in Belo Horizonte. No scientific criteria were used to locate the bases, and they

Fig. 1 EMS demand in Belo Horizonte city

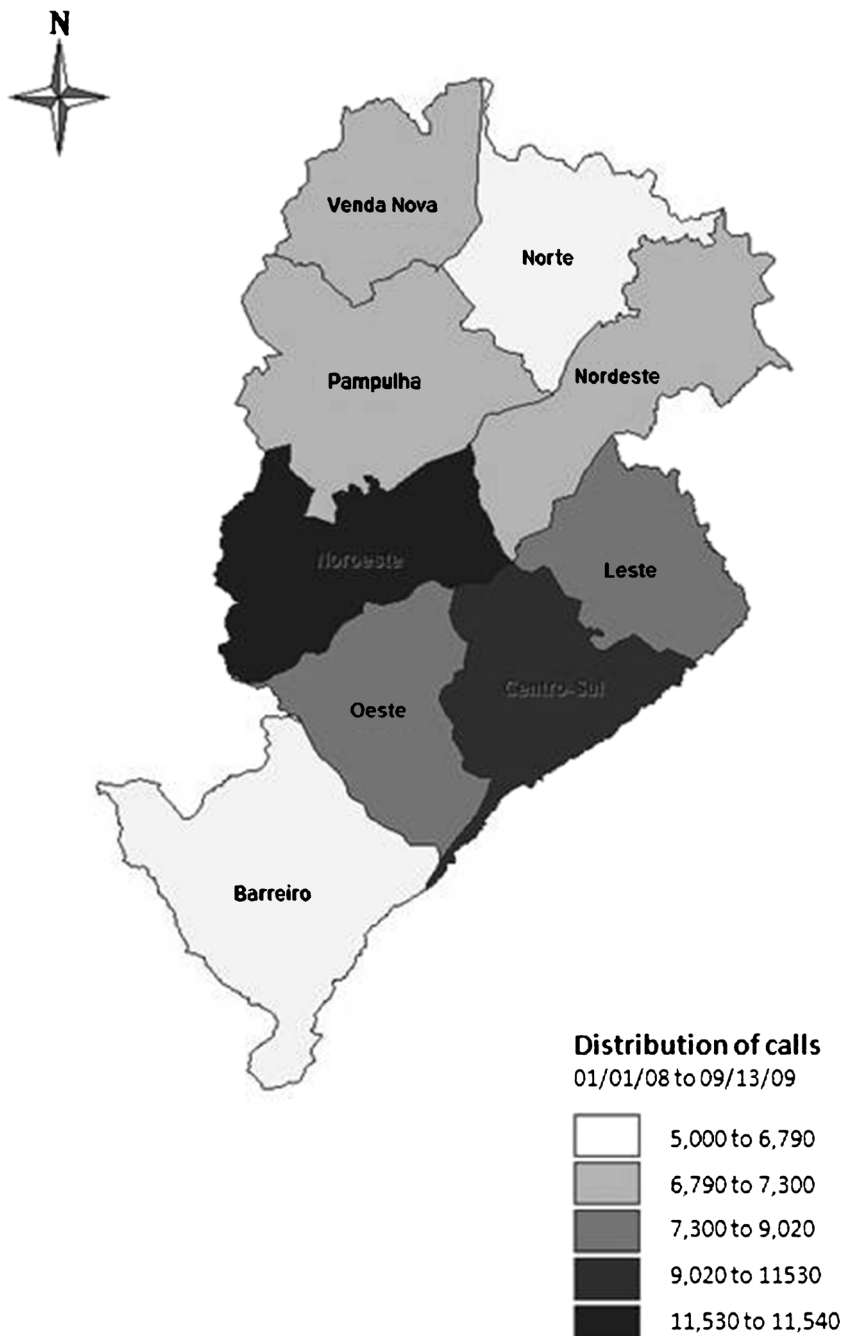


Table 2 Distribution of calls (01/01/08 to 09/13/09)

Distribution of calls		
Morning	20,296	29 %
Afternoon	20,090	28 %
Evening	19,349	27 %
Dawn	9,512	13 %
Not registered	1,459	2%
Total	70,706	100 %

were installed in public buildings or others available places. Table 3 shows data for these ambulance bases and their UTM (Universal Transverse Mercator) coordinates.

There is at least one ambulance base in each administrative region of Belo Horizonte, which has nine administrative regions. Typically, there is only one type of ambulance per base, except for bases 3 and 5. Bases 11, 12, and 19 each have two allocated ambulances.

3.3 Hospital locations

Belo Horizonte has 15 hospitals and basic health care units (identified as UPAs). A UPA is smaller than a hospital and does not have the same facilities. Table 4 shows data for these facilities.

3.4 Demand points

The demand points are randomly spread throughout the entire city. To simplify the model, the demand of each region was centralized. Figure 2 shows the demand points for the city. The model uses the UTM coordinates as central points in each area.

3.5 Distance between the points of interest

The model used the UTM coordinates of the 71 demand points, 22 ambulance bases, and 15 hospitals to calculate the travel times based on the Euclidean distance between each. The model used a factor to correct these Euclidean distances to approximate the real distance traveled by an ambulance. A factor of 1.366 was used for urban areas as shown in the study of São Paulo, Brazil [33].

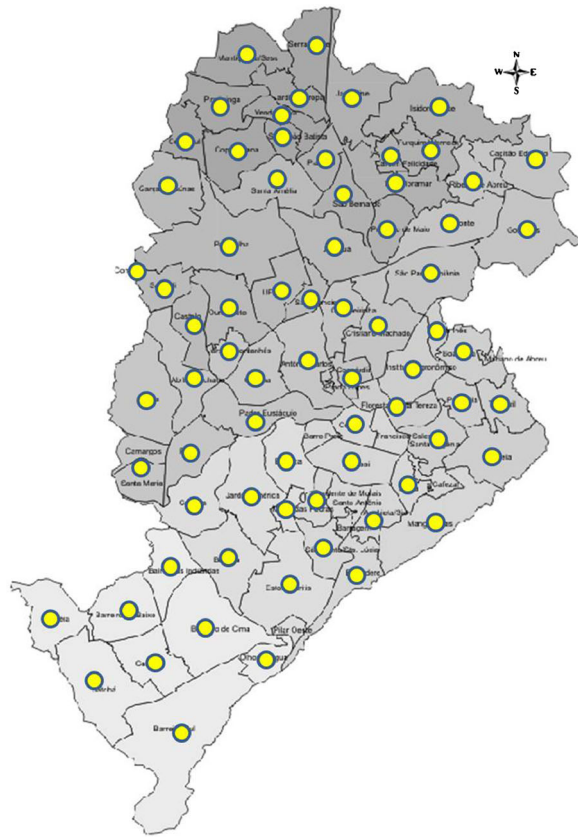
Table 3 Ambulance bases

Base	Basic ambulance	Advanced ambulance	Region	UTM coordinates	
				X	Y
1	—	1	Pampulha	608887.03	7,803,872.68
2	—	1	Nordeste	610288.82	7,798,748.39
3	1	1	Venda Nova	610,267.00	7,808,237.4
4	—	1	Centro Sul	611,687.50	7,795,581.79
5	1	1	Oeste	605,601.01	7,792,252.49
6	1	—	Venda Nova	609,015.67	7,808,665.39
7	1	—	Venda Nova	609,636.86	7,808,116.69
8	1	—	Nordeste	613,241.83	7,802,407.69
9	1	—	Noroeste	607,816.43	7,797,622.15
10	1	—	Centro Sul	610,402.96	7,791,325.68
11	2	—	Leste	614,481.28	7,798,990.56
12	2	—	Centro Sul	610,855.97	7,797,518.67
13	1	—	Leste	611,821.57	7,797,705.74
14	1	—	Barreiro	603,253.19	7,791,095.84
15	1	—	Norte	611,598.04	7,804,814.99
16	1	—	Centro Sul	612,314.1	7,796,592.16
17	1	—	Pampulha	604,483.2	7,802,482.64
18	1	—	Centro Sul	612,471.63	7,796,753.97
19	2	—	Oeste	608,882.58	7,796,808.42
20	1	—	Barreiro	602,739.96	7,788,778.63
21	1	—	Santa Luzia	618,428.72	7,814,553.46
22	1	—	Santa Luzia	611,166.52	7,811,706.68

Table 4 Hospital location

Hospital/UPA	UTM coordinates	
	UTM X	UTM Y
1 UPA Barreiro	602,739.86	7,788,778.74
2 UPA Oeste	607,831.80	7,793,930.64
3 UPA Centro-Sul	612,471.63	7,796,754.08
4 UPA Leste	614,441.80	7,797,973.79
5 UPA Norte	612,298.59	7,803,342.19
6 UPA Venda Nova	609,592.35	7,808,252.32
7 UPA Pampulha	604,483.20	7,802,482.64
8 UPA Nordeste	611,977.62	7,799,775.71
9 Hospital das Clínicas	611,767.96	7,796,253.31
10 Hospital João XXIII	611,895.62	7,796,328.76
11 Hospital Odilon Behrens	610,298.66	7,798,654.56
12 Hospital Risoleta Tolentino Neves	610,267.00	7,808,237.29
13 Hospital Julia Kubitschek	605,555.00	7,789,493.26
14 Hospital Alberto Cavalcanti	605,728.20	7,797,968.28
15 Hospital Santa Casa	612,315.29	7,796,597.80

Fig. 2 Demand point locations in Belo Horizonte city



71 demand points

3.6 Costs

Data from the Brazil National Health System (available at www.datasus.gov.br) for the year 2010 are presented in this study.

The activation operation cost (CI) is the sum of expenses for installing an ambulance base. The average cost in Belo Horizonte is 162,790.69 US dollars.

The average ambulance acquisition cost (CW) is the average cost to purchase an ambulance. We considered the average cost to be 82,938.73 US dollars for both basic and advanced ambulances to simplify the model.

The average transportation cost (CT) is the sum of all expenses related to ambulance travel (paramedic team, fuel, tires.), and maintenance. We considered the average CT to be 23.96 US dollars per trip.

3.7 Facility location modeling

3.7.1 Sets

- I demand points
- J ambulance bases

- H hospitals
- W ambulance types

3.7.2 Parameters

- TT total time (in minutes) per period, t , (morning, afternoon, evening, or dawn)
- TS_{ji} the minimum time (in minutes) between the ambulance base, j , and demand point, i
- TH_{ih} the minimum time (in minutes) between demand point, i , and hospital, h
- TB_{hj} the minimum time (in minutes) between hospital, h , and ambulance base, j
- d_{iw} the need for an ambulance of type, w , at demand point, i
- CI base activation cost
- CW cost for each ambulance
- CT transportation cost
- Q_{jw} maximum number of ambulances of type, w , at base, j
- DF ambulance availability (85 %)
- NA_w total number of ambulances of type, w , in operation
- TR response time (minutes)
- n number of demand points
- p number of candidate ambulance bases

k maximum number of ambulance bases to be activated
 u number of hospitals

3.7.3 Variables

x_{ijhw} total number of trips an ambulances of type, w , makes from base, j , to demand point, i , and hospital, h
 y_j binary variable ($y_j = 1$ if base, j , is activate and $y_j = 0$ otherwise)
 A_{jw} number of ambulances of type, w , allocated to base, j

3.7.4 Model

$$\text{Min} \sum_{j=1}^p (CI * y_j) + \sum_{j=1}^p \sum_{w=1}^2 (CW * A_{jw}) + \sum_{i=1}^n \left(\sum_{j=1}^p \sum_{h=1}^u \sum_{w=1}^2 (CT * (TS_{ji} + TH_{ih} + TB_{hj})) * x_{ijhw} \right) \quad (1)$$

s.t.

$$\sum_{j=1}^p \sum_{h=1}^u x_{ijhw} \geq d_{iw} \quad \forall i, w \quad (2)$$

$$A_{jw} \leq Q_{jw} * y_j \quad \forall j, w \quad (3)$$

$$\sum_{i=1}^n \sum_{h=1}^u x_{ijhw} * (TS_{ji} + TH_{ih} + TB_{hj}) \leq (TT_i * DF * A_{jw}) \quad \forall j, w \quad (4)$$

$$\sum_{j=1}^p y_j \leq k \quad (5)$$

$$\sum_{i=1}^n \sum_{h=1}^u x_{ijhw} \leq \sum_{i=1}^n d_{iw} * y_j \quad \forall j, w \quad (6)$$

$$\sum_{j=1}^p A_{jw} = NA_w \quad \forall w \quad (7)$$

$$x_{ijhw} \in Z^+ \quad \forall i, j, h, w \quad (8)$$

$$y_j \in \{0, 1\} \quad \forall j \quad (9)$$

$$A_{jw} \in Z^+ \quad \forall j, w \quad (10)$$

The objective function (1) minimizes the total costs. The constraint (2) states that every point of demand must be

served. Constraint (3) states the maximum number of ambulances per base. Constraint (4) requires that the total travel time to be less than or equal to the availability. Constraint (5) restricts the number of installed bases. Constraint (6) states that only active bases can be chosen to serve a demand, i . Constraint (7) establishes the total number of ambulances installed in the system. Constraints (8) through (10) define the decision variable types. This optimization model was implemented using AMPL/CPLEX.

3.8 Discrete event simulation modeling

The discrete event simulation (DES) model used in this study was adapted from Silva and Pinto [6], and the computer model was implemented in Arena 11.0. The simulation model was used to compare the response time generated in the optimization model to that of the simulation.

The main changes in the original model inputs are intended to i) increase the number of ambulances in the system, ii) create new ambulance bases, iii) reallocate ambulance bases, and iv) create new hospitals. Figure 3 provides a basic description of the simulation model.

The rescue process begins with an incoming emergency call at central, which is answered by one of the attendants. Before initializing the required procedures, the model first determines the call characteristics: its geographic origin, type (clinical, trauma or psychological), medical conduct (dispatch an ambulance or provide direction by phone) and the UTM coordinates as described in Section 3.

The attendant proceeds with the initial call analysis and trying to identify its nature (solicit information and identify hoax calls, wrong calls or medical assistance solicitation). Finally, if the call is an emergency, the initial information, such as incident location and victim data, is collected. The physician in charge of the regulation then analyzes its severity and decides whether to dispatch an ambulance.

This model considers two basic pieces of information to determine whether an ambulance should be dispatched: the UTM coordinates from which the ambulance will be released (ready for serving a new demand) and the traveling time to the incident. Considering both factors guarantees the model dispatches the ambulance that will arrive on scene soonest and not just the closest ambulance. At the scene, the medical team delivers initial treatments, and in more severe cases, the patient is delivered to the nearest hospital.

After delivering the patient, the medical team must determine the need for material replacement and/or maintenance. In this case, the ambulance is available to serve a new call after such activities are complete.

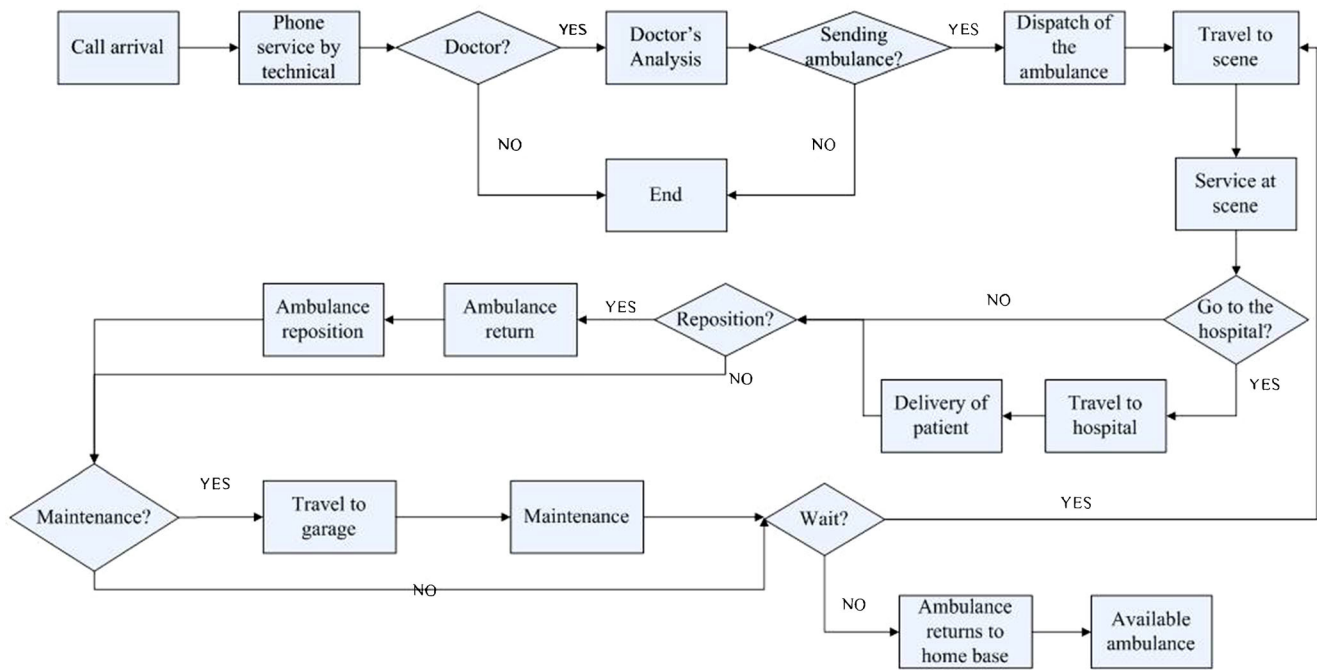


Fig. 3 Flowchart for answering emergency calls

If the ambulance is already required for a new service, the medical team proceeds directly to the incident. Otherwise, the ambulance returns to its base.

4 Scenario analyses and computer results

Assuming optimizing the ambulance base positions may reduce the response time, we proposed six scenarios for the optimization model. Furthermore, we used the best results as the simulation model inputs to determine the dynamic system behavior.

Scenarios 1, 2 and 3 represent systems with three different ambulance base positions. For these scenarios, the number of ambulances in the system was the number operating in 2010 (20 basic and 5 advanced ambulances) with the same number of bases (22 bases). The purpose of these scenarios was to determine the relationship between the location and response time for each system.

Scenario 1—The model uses the exact configuration of the 2010 system (base locations and ambulance numbers). Scenario 2—The model arbitrarily chosen the configuration of the 22 bases. Scenario 3—The model uses a configuration for the 22 bases chosen according to the proportional demand (locations with higher demand have more installed bases).

Scenario 4 uses the current system configuration, i.e., 22 ambulance bases and 27 ambulances (22 basic and 5 advanced). This scenario was also used to calculate the shortest response time.

Scenarios 5 and 6 represent different characteristics. We insert more candidate bases throughout the city. For both scenarios, we used 71 candidate bases. The difference between scenario 5 and 6 is that scenario 5 may activate up to 22 bases, while scenario 6 activates exactly 22 bases.

Each scenario used several time schedules (morning, afternoon, evening and dawn) with a distinct vehicle speed, ambulance number at each base (2, 3 and 5 ambulances) and required response time (from 21 min, the current system time, to 8 min, which is the response time suggested by the World Health Organization).

We analyzed more scenarios but do not show them because they were infeasible or yielded worse solutions than previously obtained. Table 5 summarizes the six scenarios used in this experiment.

The analytical results for scenarios 1, 2, 3, and 4 indicate maintaining the current operating base locations is not the best strategy. Therefore, the hypothesis that the operating base locations may reduce the response time will be tested using scenarios 5 and 6.

Based on this hypothesis, the new ambulance base locations were proposed in scenarios 5 and 6 to reduce the service response time, especially during the morning and afternoon. Table 5 shows that the response times improve relative to those for scenarios 1, 2, 3, and 4.

To analyze the configurations generated for the proposed scenarios in detail, Tables 6, 7, 8, and 9 summarize the results for a specific period. Scenarios 2 and 3 were excluded because they yielded no response below 21 min. Notably, no seasonal

Table 5 Feasible (1) and infeasible (0) scenarios

		06:00 AM–12:00 PM						12:00 PM–06:00 PM						06:00 PM–12:00 AM						12:00 AM–06:00 AM					
Scenário/Response time (min)		1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6	1	2	3	4	5	6
Up to 2 ambulances per base	21	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	20	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	19	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	17	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	0	1	1	1
	15	0	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1
	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1
	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Up to 3 ambulances per base	21	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	20	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	19	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	17	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	0	1	1	1
	15	0	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1
	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1
	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Up to 5 ambulances per base	21	1	0	0	1	1	1	1	0	0	1	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	20	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	19	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	1	1	1	1
	17	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1	1	0	0	1	1	1
	15	0	0	0	0	1	1	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	1
	13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1
	11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

weather changes significant altered the EMS operations, i.e., the system behaves the same throughout the year.

Table 6 shows that, during dawn (12:00 AM to 06:00 AM), scenario 6 yielded feasible solutions with a

Table 6 Scenarios 12:00 AM–06:00 AM

Scenario	Max ambulance per base (Q)	Period	Better Response Time (TR)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)
1	2	12:00 AM–06:00 AM	13'	10	20	5
4	2	12:00 AM–06:00 AM	13'	11	22	5
5	2	12:00 AM–06:00 AM	11'	11	22	5
6	2	12:00 AM–06:00 AM	8'	22	22	5
1	3	12:00 AM–06:00 AM	13'	7	20	5
4	3	12:00 AM–06:00 AM	13'	8	22	5
5	3	12:00 AM–06:00 AM	11'	8	22	5
6	3	12:00 AM–06:00 AM	8'	22	22	5
1	5	12:00 AM–06:00 AM	13'	4	20	5
4	5	12:00 AM–06:00 AM	13'	5	22	5
5	5	12:00 AM–06:00 AM	11'	5	22	5
6	5	12:00 AM–06:00 AM	8'	22	22	5

Table 7 Scenarios 06:00 AM–12:00 PM

Scenario	Max ambulance per base (Q)	Period	Better Response Time (TR)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)
1	2	06:00 AM–12:00 PM	21'	10	20	5
4	2	06:00 AM–12:00 PM	21'	11	22	5
5	2	06:00 AM–12:00 PM	15'	11	22	5
6	2	06:00 AM–12:00 PM	15'	22	22	5
1	3	06:00 AM–12:00 PM	21'	7	20	5
4	3	06:00 AM–12:00 PM	21'	8	22	5
5	3	06:00 AM–12:00 PM	15'	8	22	5
6	3	06:00 AM–12:00 PM	15'	22	22	5
1	5	06:00 AM–12:00 PM	21'	4	20	5
4	5	06:00 AM–12:00 PM	21'	5	22	5
5	5	06:00 AM–12:00 PM	15'	5	22	5
6	5	06:00 AM–12:00 PM	15'	22	22	5

response time of 8 min regardless of the maximum number of ambulance bases. Note that this scenario used 22 ambulance bases and 27 ambulances.

Tables 7 and 8 show the morning (06:00 AM to 12:00 PM) and afternoon (12:00 PM to 06:00 PM) periods, respectively. The response times obtained for scenarios 5 and 6 (15 min) represent a considerable reduction relative to scenarios 1 and 4 (21 min).

Table 9 shows the last period (06:00 PM to 12:00 AM). Scenarios 5 and 6 both achieved a response time of 13 min using 27 ambulances.

These optimization results indicate scenarios 5 and 6 significantly reduce the response time. Even though scenario 5 has a smaller number of installed ambulance bases, it exhibits a similar response time to scenario 6.

Because scenarios 5 and 6 use new ambulance bases and significantly reduce the response time, we decided to implement the optimizer-generated configurations in the DES model.

This model analyzes current and future scenarios to plan strategic actions and manage the system. For instance, the model can analyze the effect installing a new hospital in the city has on the response time. It seems intuitive that the hospital and basic care unit locations affect the response time. To analyze this influence, two new hospitals were included at distinct points within the city. Table 10 shows the results of this analysis. Despite the increased number of hospitals, the response time remained above 15 min during peak hours due to the slow ambulances speed and low ambulance number. Furthermore, the elapsed time

Table 8 Scenarios 12:00 PM–06:00 PM

Scenario	Max ambulance per base (Q)	Period	Better Response Time (TR)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)
1	2	12:00 PM–06:00 PM	21'	10	20	5
4	2	12:00 PM–06:00 PM	21'	11	22	5
5	2	12:00 PM–06:00 PM	15'	11	22	5
6	2	12:00 PM–06:00 PM	15'	22	22	5
1	3	12:00 PM–06:00 PM	21'	7	20	5
4	3	12:00 PM–06:00 PM	21'	8	22	5
5	3	12:00 PM–06:00 PM	15'	8	22	5
6	3	12:00 PM–06:00 PM	15'	22	22	5
1	5	12:00 PM–06:00 PM	21'	4	20	5
4	5	12:00 PM–06:00 PM	21'	5	22	5
5	5	12:00 PM–06:00 PM	15'	5	22	5
6	5	12:00 PM–06:00 PM	15'	22	22	5

Table 9 Scenarios 06:00 PM–12:00 AM

Scenario	Max ambulance per base (Q)	Period	Better Response Time (TR)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)
1	2	06:00 PM–12:00 AM	17'	10	20	5
4	2	06:00 PM–12:00 AM	17'	11	22	5
5	2	06:00 PM–12:00 AM	13'	11	22	5
6	2	06:00 PM–12:00 AM	13'	22	22	5
1	3	06:00 PM–12:00 AM	17'	7	20	5
4	3	06:00 PM–12:00 AM	17'	8	22	5
5	3	06:00 PM–12:00 AM	13'	8	22	5
6	3	06:00 PM–12:00 AM	13'	22	22	5
1	5	06:00 PM–12:00 AM	17'	4	20	5
4	5	06:00 PM–12:00 AM	17'	5	22	5
5	5	06:00 PM–12:00 AM	13'	5	22	5
6	5	06:00 PM–12:00 AM	13'	22	22	5

between the call arrival and ambulance dispatch was too high, approximately 5 min, for the Belo Horizonte EMS. The intricate city design leads to heavy traffic during peak hours and drastically reduces the average ambulance speeds, which is crucial for the response time.

Therefore, the next step is to check whether the results using the optimization model responses converge on the simulation model. The ambulance base locations and ambulance allocation was set using the optimization model. The simulation model analyzed the dynamic behavior of the system to evaluate the configuration proposed by the optimization model. The simulation only used the optimization model solutions generated for the morning period because this period had an increased demand with greater system utilization.

Table 11 shows the maximum number of ambulances per base, the number of activated bases, the number of basic and advanced units, and the average response times using the optimization and simulation models.

The gaps between the response times for the optimization and simulation models were supposedly due to the random nature of the process and the time spent by operators and/or technicians at the call center, i.e., the time required to dispatch the ambulance. The optimization model assumes these tasks require 2 min, while the simulation model had an average process time of approximately 5 min.

Furthermore, the optimization model uses distinct average ambulance speeds and demand points (71 points) for morning, daytime, nighttime, and dawn. The simulation model assumes the average ambulance speed was 25 km/h between 8:00 AM and 8:00 PM and 50 km/h between 8:00 PM and 8:00 AM, with only nine demand points.

Scenario 5 yielded a 15 min response time using only 5 activated bases. This fact might be relevant from an economic viewpoint. Another interesting fact is the inverse relationship between the number of bases and response time for the simulation model. We did not observe this phenomenon in the optimization model.

Table 10 Solution for an increased number of hospitals (Scenarios 7 and 8)

Scenario	Max ambulance per base (Q)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)	Response time obtained by optimization (min)	Response time obtained by simulation (min) CI 95 %
7	2	22	22	5	15	17.33+−0.8
	3	22	22	5	15	19.01+−0.8
	5	22	22	5	15	19.76+−0.8
8	2	22	22	5	15	16.71+−0.8
	3	22	22	5	15	16.88+−0.8
	5	22	22	5	15	16.71+−0.8

Table 11 Best solutions (Scenarios 5 and 6)

Scenario	Max ambulance per base (Q)	Installed bases (j)	Total ambulance (Basic)	Total ambulance (Advanced)	Response time obtained by optimization (min)	Response time obtained by simulation (min) CI 95 %
5	2	11	22	5	15	17.45+−0.8
	3	7	22	5	15	19.08+−0.8
	5	5	22	5	15	19.96+−0.8
6	5	22	22	5	15	16.75+−0.8
	3	22	22	5	15	16.92+−0.8
	5	22	22	5	15	16.92+−0.8

Obviously, this study is only preliminary and we should assess the configurations for scenarios 5 and 6 in terms of their operational capabilities and economics before changing the real system.

5 Conclusions

After studying the scenarios and analyzing the most feasible solutions, we concluded the ambulance number and base locations affect the EMS performance. However, buying ambulances and activating new bases does not necessarily improve the system performance.

A balanced number of ambulances allocated across certain bases at certain times improved the system performance without increasing the total number of ambulances, i.e., just changing the ambulance allocation obtained WHO targets.

The elapsed time between the call arrival and ambulance dispatch was too large (approximately 4.5–5.4 min) in Belo Horizonte. Reducing this time should reduce the response time.

Another important issue is the base distribution. Currently, the system operates with essentially one basic unit at each base. This study shows that the number of ambulances at each base can be increased to decrease the number of activated bases. This action may decrease the costs without reducing performance. To accomplish this feat, managers should prepare strategic bases for operation and enlarge the parking areas.

This study also showed the importance of daily seasonality on the service quality. Future studies may further reduce the analytical time range by considering, for example, the hourly demand variations. Furthermore, monthly demand variations can be included. Such a study would create a more specific and appropriate service solution.

Others possible alternatives include using intermittently operating bases where a smaller number of ambulance bases operate during evening and dawn when the service demand is significantly decreased. Such a system should lower operating and maintenance costs.

Finally, this study shows how to join two operational research techniques, optimization and simulation, to analyze the dynamic EMS behavior instead of using empirical rules and procedures.

Conflict of interest The authors declare they have no conflicts of interest.

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